

Ibrahim O. Jimoh

Artificial Intelligence & Robotics

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RESEARCH INTERESTS

Developing intelligent robotic systems that can perceive, learn, and interact appropriately in diverse cultural and social contexts. Human-robot interaction, robot learning and perception, robot manipulation, culturally sensitive social robotics, reinforcement learning, deep learning, and AI safety.

EDUCATION

Master of Science, Engineering Artificial Intelligence | July 2023 – May 2025

Carnegie Mellon University (CMU) | Pittsburgh, USA

Specializations:

- Machine Learning, Deep Learning, Robotics, Large Language Models, Reinforcement Learning

Relevant Coursework:

- Applied Stochastic Processes, Machine Learning for Engineers, Introduction to Deep Learning, Cognitive Robotics, Data Inference and Applied Machine Learning, AI System Design, Principles and Applications of AI, AI/ML Operations, Humanoid Robotics and Cognition, Research Methods

Bachelor of Engineering, Computer Engineering | September 2015 – September 2021

Federal University of Technology (FUT) | Minna, Nigeria

Thesis: Development of a Drowsiness Detection System for Road Accident Reduction and Control using Computer Vision Techniques

RESEARCH EXPERIENCE

Research Associate | June 2025 – Present

AI & Robotics Laboratory, Culturally Sensitive Social Robotics | CMU

- Developing a comprehensive cultural knowledge base structured into a cultural ontology to guide robot interactions in culturally attuned manners across diverse African contexts.
- Implementing advanced robot localization software using visual landmarks, triangulation, and trilateration methods to enable precise spatial awareness for social robot navigation and interaction.

Research Assistant | June 2024 – May 2025

AI & Robotics Laboratory, Culturally Sensitive Social Robotics | CMU

- Conducted ethnographic research to identify culturally sensitive robot behaviors and interaction patterns socially acceptable in African contexts, informing the design of culturally-aware robotic systems.
- Developed and implemented a Behavior Tree Domain Specific Language (DSL) for structured human-robot interaction scenarios, enabling flexible and maintainable robot behavior design.
- Created multi-modal robot localization system combining odometry, IMU data, ArUco marker detection, and YOLO-based landmark recognition for robust pose estimation in dynamic environments.

Research Assistant | January 2021 – September 2021

Computer Engineering Laboratory | Federal University of Technology

- Investigated computer vision algorithms for vehicle safety systems, conducting comprehensive surveys on vehicular accident statistics and causal factors.
- Developed in-vehicle drowsiness detection system using Histogram of Oriented Gradient (HOG) facial predictors and eye aspect ratio monitoring with real-time alarm capabilities and emergency notification features.

PUBLICATIONS

- A. Akinade, D. Barros, E. Birhan, M. Danso, Y. Haile, **I. Jimoh**, C. Osano, P. Ranchod, M. Richard, B. Rosman, B. Shimelis, T. Tefferi, and D. Vernon, "Culturally Sensitive Social Robotics for Africa," chapter in Cultural Robotics Book: Diversified Sustainable Practices, Springer Nature [To be published].
- I. Jimoh**, H. Equbay, C. Osano, T. Tefferi, and D. Vernon, "Behavior Trees for Culturally Sensitive Social Robots: African Culture Case Study," Abstract in Robotics in Africa Forum at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
- A. Akinade, D. Barros, M. Danso, Y. Haile, E. Birhan, B. Shimelis, C. Osano, P. Ranchod, M. Richard, B. Rosman, **I. Jimoh**, T. Tefferi, and D. Vernon, "Culturally Sensitive Social Robotics for Africa," Poster Presentation in Cultural Robotics: Diversified Sustainable Practices, Workshop at IEEE/ACM Human Robot Interaction (HRI) Conference, 2025.

- Y. U. Murph, A. Avoka, **I. Jimoh**, and F. Rurangwa, "Breaking Mental Health Silence: Men from Different Cultures and Generations," Case Studies in International Education, Vol. 5 No. 2, Dec. 2024.

WORK EXPERIENCE

Software Engineer Intern | January 2023 – June 2023

Optimus Bank | Lagos, Nigeria

- Led as a project lead, coordinating timelines, ensuring deliverable completion, and fostering collaboration between development and business units for critical AI integration projects.
- Served in a scrum master capacity, facilitating daily stand-ups, sprint planning, and retrospectives, resulting in significant improvements in team productivity and task completion rates.
- Designed and developed RESTful APIs for customer support virtual assistant, enabling seamless integration with existing banking infrastructure and improving service efficiency.
- Maintained rigorous data security protocols and ensured 100% regulatory compliance with banking security standards and industry requirements.

IT Engineer Intern | January 2022 – December 2022

Protogy Global Services Limited | Ibadan, Nigeria

- Managed and maintained electrical meter programming software, implementing updates and optimizing utility performance across production systems.
- Enhanced production efficiency by over 60% through development of Python-based inventory management application, streamlining product entry workflows.
- Resolved critical software and hardware issues in meter pipeline systems, ensuring operational continuity and product quality.
- Recognized as the most efficient and best performing intern for outstanding contributions to operational improvements.

Hardware and Software Technician | May 2016 – May 2019

Canaanland Technologies | Ilorin, Nigeria

- Maintained, troubleshooted, and repaired computer systems and devices for diverse corporate and individual clients, ensuring high customer satisfaction.
- Provided comprehensive technical support and customized software solutions tailored to client needs and specifications.
- Developed strong customer relations skills through direct client interactions and technical consultation services.

SELECTED RESEARCH PROJECTS

Deep Reinforcement Learning for Mobile Robot Social Navigation | Sep 2024 – Dec 2024

- Developed deep RL framework jointly modeling human-robot and human-human interactions in dynamic social spaces.
- Implemented self-attention architecture capturing long-range dependencies for complex crowd interaction modeling.

LLM-Based Investment Advisor using RAG | Sep 2024 – Dec 2024

- Built scalable LLM-powered platform democratizing financial advising with risk profiling, asset allocation, portfolio forecasting, and personalized recommendations using Retrieval-Augmented Generation.

Deepfake Voice Detection for Fraud Prevention | Jun 2024 – Aug 2024

- Developed and trained deep CNN model to distinguish authentic and synthetic voice recordings with high accuracy.

Robot Arm Manipulation for Structured Tasks | Mar 2024 – Apr 2024

- Developed control strategies for robotic arm pick-and-place tasks demonstrating precise manipulation in structured and semi-structured environments.

Synchronizing Multimodal Behaviors - Eva Robot Head | Sep 2023 – Dec 2023

- Built responsive humanoid robot head using Unity, VR, 3D printing (Fusion 360, Ultimaker Cura).
- Implemented facial expression recognition and mimicking using neural networks; enhanced open-source control code.

TEACHING & MENTORSHIP

Teaching Assistant – Robotics Principles and Practice | Fall 2025

Carnegie Mellon University College of Engineering

- Supported graduate students in robotics concepts, kinematics, motion planning, and control.
- Assisted with lab setup, grading, and detailed feedback on hands-on robotics assignments.

AI Alignment Facilitator | Aug 2025 – Oct 2025

European Network for AI Safety

- Facilitated weekly international discussion groups on AI safety and alignment as part of AI Safety Collab initiative.
- Simplified complex technical concepts and fostered collaboration on AI governance and safety topics.

TECHNICAL SKILLS

Programming Languages:

- Python, C, C++, C#, Java, JavaScript

AI & Machine Learning:

- PyTorch, TensorFlow, Neural Networks, Scikit-Learn, APIs, RAG, LangChain, Pandas, NumPy

Robotics:

- ROS, Gazebo, OpenAI Gym, Robot Kinematics, Motion Planning, Control Systems, Sim2Real

Cloud & DevOps:

- AWS, GCP, Docker, Kubernetes

Developer Tools:

- Git, GitHub, Linux/Ubuntu, Unity3D, LaTeX, Arduino, Raspberry Pi

HONORS & AWARDS

- **HRI Conference Grant Award** | ACM SIGAI, 2025 | Competitive grant to present research at HRI Conference in Melbourne, Australia
- **Mastercard Foundation Scholarship** | 2023 | Highly competitive master's degree scholarship for African students
- **Best School Representative** | International Mathematics Olympiad (IMO), 2015 | Recognized for exemplary performance

CONFERENCE PRESENTATIONS

- **IEEE/ACM HRI Conference** | Melbourne, Australia, 2025 | Poster: "Culturally Sensitive Social Robotics for Africa" at Cultural Robotics Workshop
- **IEEE/RSJ IROS Conference** | Abu Dhabi, UAE, 2024 | Abstract: "Behavior Trees for Culturally Sensitive Social Robots" at Robotics in Africa Forum
- **Touch Sensing and Processing Summer School (TSPSS)** | Dresden, Germany, 2024 | Hands-on training in tactile sensing and manipulation with Meta AI, TouchLab, Stanford, TU Dresden

LEADERSHIP & SERVICE

- **Communications Lead** | Afrisnet Graduate Committee, Jun 2024 – Present | Organizing workshops and career development sessions for 500+ African graduate scholars
- **Interim Chair & Vice Chair** | IEEE Student Branch, CMU-A, May 2024 – May 2025 | Coordinated branch activities, organized workshops, led IEEEExtreme 18.0 participation
- **Community Engagement Manager** | Research Club, CMU-A, Dec 2023 – Dec 2024 | Established faculty-student research collaboration, organized research workshops and panels
- **Volunteer** | Solid Africa, Dec 2023 | Provided aid and distributed relief materials to 100+ vulnerable hospital patients in Rwanda

PROFESSIONAL MEMBERSHIPS

- Member, Association for Computing Machinery (ACM)
- Graduate Member, Institute of Electrical and Electronics Engineers (IEEE)
- Graduate Member, IEEE Robotics and Automation Society (IEEE-RAS)
- Graduate Member, Nigeria Society of Engineers (NSE)
- Member, Google Developers Group (Active since 2020)